

DISASTER MANAGEMENT-[3-1] IMPORTANT 36 TOPIC'S:

UNIT I

1. Introduction to Disaster Management –

1. Disaster Management (DM) is a systematic process to prepare, prevent, respond, and recover from disasters.
 2. It aims to minimize loss of life, property, and environmental damage.
 3. DM works through four phases: Preparedness, Mitigation, Response, Recovery.
 4. It involves multi-agency coordination between government, NGOs, military, and community.
 5. DM ensures community resilience by strengthening coping capacity.
 6. A proactive DM system reduces panic through awareness and training.
 7. DM includes planning for all hazards: natural, man-made, biological, and technological.
 8. Scientific tools like GIS, IoT sensors, and satellite monitoring enhance DM effectiveness.
 9. DM also focuses on post-disaster rehabilitation and future risk reduction.
 10. Overall, DM transforms a reactive society into a prepared and resilient one.
- 2. Key Concepts – Disaster, Hazard, Vulnerability, Risk, Capacity – 10 Points**
1. A **Disaster** is a serious disruption causing large-scale loss beyond local capacity.
 2. A **Hazard** is a potential threat (earthquake, cyclone, chemical leak).
 3. **Vulnerability** refers to conditions that make people or property susceptible to harm.
 4. **Risk = Hazard × Vulnerability**; high vulnerability increases disaster impact.
 5. **Capacity** is the strength and resources available to reduce disaster risk.
 6. Hazards become disasters only when vulnerabilities are high.
 7. Socio-economic factors like poverty increase vulnerability.
 8. Structural capacity includes safe buildings; non-structural includes knowledge.
 9. DM aims to reduce vulnerability and increase capacity.
 10. Understanding these concepts helps in designing effective DM plans.

Mini-Case-Study:

In the **2010 Haiti earthquake**, weak buildings + poverty (high vulnerability) resulted in 230,000

deaths, despite similar hazards occurring elsewhere with fewer casualties.

3. Impacts of Disasters – Social, Economic, Environmental – 10 Points

1. Disasters cause large-scale displacement affecting shelter and living conditions.
2. Social systems break down—education, healthcare, transport collapse.
3. Psychological impacts include trauma, fear, and stress.
4. Economically, disasters damage industries, roads, bridges, and farms.
5. Recovery costs run into billions; national GDP may decline.
6. Local livelihoods like farming and fishing are disrupted.
7. Environment suffers through soil erosion, pollution, and habitat destruction.
8. Floods, landslides, and fires change landforms permanently.
9. Waste generation increases, causing secondary hazards.
10. Long-term impacts include poverty, migration, and ecosystem imbalance.

CaseStudy:KeralaFloods2018

Mass displacement, ₹31,000 crore loss, landslides, and river changes reflect all three impacts.

4. Prevention and Mitigation Strategies – 10 Points

1. **Prevention** includes avoiding hazard-prone areas and banning construction on riverbanks.
2. **Mitigation** focuses on reducing disaster impact (strong buildings, embankments).
3. Use of early warning systems helps prepare communities in advance.
4. Land-use planning reduces exposure of people to hazards.
5. Afforestation and watershed development prevent floods and drought.
6. Retrofitting old buildings reduces earthquake damage.
7. Community awareness reduces panic and increases preparedness.
8. Insurance transfers financial risk.
9. Enforcing safety laws in industries prevents man-made disasters.
10. Overall, mitigation reduces long-term risk and saves lives.

Case-Study:

Japan's earthquake-resistant buildings prevented massive deaths in the 2011 earthquake.

5. Disaster Management Act, 2005 – 10 Points

1. Provides India a legal framework for disaster preparedness and response.
2. Established **NDMA** under the Prime Minister for national policy making.
3. Formed **SDMAs** headed by Chief Ministers at state level.
4. Created **DDMAs** for district-level implementation.
5. Defines powers, responsibilities, and coordination mechanisms.
6. Mandates preparation of disaster management plans at all levels.
7. Established **NDRF** as a specialized force for rescue operations.
8. Provides financial mechanisms like NDRF/SDRF funds.
9. Ensures community participation and awareness programs.
10. Used during national emergencies like **COVID-19 lockdown** for legal enforcement.

6. Institutional Framework for Disaster Management in India – 10 Points

1. **NDMA**: National level policy formation, guidelines, coordination.
2. **NDRF**: Specialized force trained in floods, quakes, chemical emergencies.
3. **SDMA**: Prepares state DM plans and implements policies locally.
4. **DDMA**: Carries out district-level rescue, relief, and evacuation.
5. **IMD**: Predicts cyclones, rainfall, and weather hazards.
6. **ISRO**: Provides satellite-based monitoring for floods & cyclones.
7. **CWC**: Monitors river water levels and flood forecasting.
8. **Ministry of Home Affairs**: Nodal ministry for DM.
9. **NGOs** assist in relief, shelters, and community rehabilitation.
10. **Panchayats** form the local response mechanism at village level.

Case-Study:

NDRF rescued thousands during the 2013 Uttarakhand floods through airlift operations and rope rescues.

7. Role of Government & International Organizations in DM – 10 Points

1. Government ensures policy, funding, training, and infrastructure for DM.
2. Coordinates rescue operations using NDRF, Army, Police, and medical teams.
3. Implements early warning systems through IMD & NDMA.
4. Provides relief camps, food, water, and rehabilitation packages.
5. International organizations support technical guidance and global standards.
6. **UNDRR** promotes disaster risk reduction strategies worldwide.
7. **WHO** provides medical teams and health guidelines during epidemics.
8. **UNICEF** protects children and supports sanitation & schooling.
9. **Red Cross** supplies emergency kits, shelters, and volunteers.
10. Joint operations improve efficiency and reduce global disaster losses.

Case-Study:

During the 2015 Nepal earthquake, India + Red Cross + WHO jointly ran rescue and medical missions.

UNIT-II –NATURAL DISASTERS

8. Definition and Nature of Natural Disasters

1. Natural disasters are extreme natural events that cause sudden and widespread destruction.
2. They occur due to natural processes such as geological, hydrological, or meteorological changes.
3. They include earthquakes, floods, cyclones, droughts, volcanoes, tsunamis, etc.
4. Natural disasters cannot be prevented but their impacts can be reduced.
5. They have high unpredictability and may occur without warning (e.g., earthquakes).
6. The severity depends on magnitude, duration, location, and vulnerability of population.
7. They can disrupt social life, economy, infrastructure, and environment.
8. Climate change is increasing frequency and intensity of many natural disasters.
9. Preparedness and mitigation measures help reduce the scale of damage.
10. Natural disasters highlight the need for strong disaster management systems.

Case-Study:

The **Nepal earthquake (2015)** demonstrated the sudden and destructive nature of natural disasters—over 9,000 deaths and collapse of historic structures.

9. Floods – Causes, Effects, and Management

1. Floods occur when water overflows onto normally dry land due to heavy rainfall or river overflow.
2. Causes include intense monsoons, deforestation, urbanization, dam failure, and poor drainage.
3. Climate change is causing unpredictable rainfall patterns, worsening floods.
4. Floods result in loss of human life, livestock, and destruction of homes.
5. Secondary impacts include waterborne diseases and displacement of communities.
6. Floods damage crops, transportation networks, and critical infrastructure.
7. Management includes building embankments, dams, and improving drainage systems.
8. Early warning systems and weather forecasting reduce casualties.
9. Floodplain zoning restricts risky construction along riverbanks.
10. Community preparedness and evacuation drills improve survival chances.

Case-Study:

The **Kerala Floods 2018** resulted from record-breaking rainfall; effective rescue operations saved lives but the economic loss exceeded ₹31,000 crore.

10. Drought – Causes, Consequences, and Mitigation –

1. Drought is prolonged deficiency of rainfall leading to water scarcity.
2. Major causes include monsoon failure, high temperature, deforestation, and overuse of water.
3. Poor irrigation systems and groundwater over-extraction worsen drought.
4. Consequences include crop failure, food insecurity, and livestock deaths.
5. Drought pushes farmers into debt, poverty, and migration.
6. Drinking water scarcity increases risk of diseases.
7. Hydrological drought affects rivers, lakes, and reservoirs.
8. Mitigation includes rainwater harvesting and watershed development.
9. Micro-irrigation (drip/sprinkler) reduces water wastage.
10. Government drought relief programs and crop insurance help farmers recover.

Case-Study:

The **Marathwada drought (2015–16)** caused severe water scarcity; tankers supplied drinking water, and the state government initiated “Jalyukt Shivar” for watershed revival.

11. Cyclones – Formation, Impact, and Preparedness – 10 Points

1. Cyclones form over warm oceans (above 26°C) due to low pressure and moisture.
2. The Coriolis force causes rotation and intensifies the system.
3. Impacts include strong winds, storm surges, tidal flooding, and infrastructure damage.
4. Coastal areas are highly vulnerable due to population density and poor housing.
5. Cyclones disrupt power, communication, agriculture, and transportation.
6. Preparedness includes early warnings, cyclone shelters, and evacuation plans.
7. IMD uses satellites and radars for precise cyclone tracking.
8. Building codes help create cyclone-resistant housing.
9. Governments conduct mock drills and awareness programs in coastal villages.
10. NDRF teams are deployed in advance for rescue and relief operations.

Case-Study:

Cyclone Fani (2019)—Odisha evacuated 1.2 million people; UN praised Odisha for “global best practice in cyclone management.”

12. Earthquakes – Causes, Effects, and Seismic Zoning – 10 Points

1. Earthquakes occur due to sudden movement of tectonic plates along fault lines.
2. Causes include plate collision, volcanic activity, and human activities like mining.
3. Earthquakes trigger landslides, tsunamis, and fires due to gas pipeline breaks.
4. They cause collapse of buildings, bridges, and critical structures.
5. Effects depend on magnitude, depth, population density, and building quality.
6. Seismic hazard mapping helps identify high-risk zones.
7. India’s seismic zones: Zone II (low) to Zone V (very high).
8. Himalayan belt and Northeast India fall under Zone V.
9. Earthquake-resistant construction and retrofitting reduce casualties.
10. Community training on “Drop–Cover–Hold” improves survival chances.

Case-Study:

The **2001 Gujarat Earthquake** (7.7 magnitude) killed 20,000 people; led to major reforms in building codes across India.

13. Tsunamis and Volcanoes – Causes and Impact – 10 Points

1. Tsunamis are large ocean waves caused by underwater earthquakes, landslides, or volcanoes.
2. They travel at high speed (800 km/h) and hit coastlines with massive force.
3. Impacts include coastal flooding, destruction of settlements, and high mortality.
4. Salinization damages agriculture for years.
5. Volcanoes erupt when magma rises through Earth’s crust.
6. Lava flows destroy infrastructure and forests.
7. Volcanic ash affects air travel and causes respiratory problems.
8. Both hazards can trigger secondary disasters like landslides.
9. Early warning systems and community evacuation plans reduce losses.
10. Hazard mapping and monitoring help predict volcanic activity.

Case-Study:

Indian Ocean Tsunami 2004 killed 2.3 lakh people globally; India established the Tsunami Early Warning Centre in Hyderabad afterward.

14. Climate Change – Global Warming, Sea-Level Rise, Ozone Depletion – 10 Points

1. Climate change refers to long-term shifts in temperature and weather patterns.
2. Global warming is caused by greenhouse gases like CO₂ and methane.
3. Burning fossil fuels, deforestation, and industrialization increase global warming.
4. Sea-level rise occurs due to melting of polar ice and thermal expansion of oceans.
5. Rising sea levels threaten coastal cities like Mumbai and Kolkata.
6. Extreme events like heatwaves, floods, and cyclones are increasing.
7. Ozone depletion is caused by CFCs released from ACs and refrigerators.
8. The ozone layer protects the Earth from harmful UV radiation.
9. Montreal Protocol successfully reduced CFC production worldwide.
10. Climate action requires renewable energy, afforestation, and carbon reduction.

CaseStudy:

Chennai Heatwave & Water Crisis (2019): Temperature exceeded 40°C; reservoirs dried up, showing impacts of climate change.

UNIT-III-MAN-MADE DISASTERS**15. Nuclear Disasters – Causes, Consequences & Safety Measures – 10 Points**

1. Nuclear disasters occur due to reactor failures, human error, natural hazards, or technical malfunction.
2. Causes include overheating of reactor core, cooling system failure, and uncontrolled chain reactions.
3. Radiation leaks lead to long-term contamination of soil, water, and air.
4. Exposure causes cancers, genetic mutation, and birth defects.
5. Large-scale evacuations are required, affecting thousands of people.
6. Economic loss includes abandoned cities, health costs, and reactor decommissioning.
7. Environmental damage persists for decades due to radioactive isotopes.
8. Safety measures include multiple cooling systems and emergency shutdown (SCRAM).
9. Strict maintenance, periodic audits, and staff training reduce risk.
10. Nuclear plants must follow international IAEA safety standards.

CaseStudy:ChernobylDisaster(1986)

Reactor explosion released massive radiation; 30 immediate deaths, long-term cancers, entire city Pripyat abandoned.

16. Chemical Disasters – Industrial Accidents & Toxic Spills – 10 Points

1. Chemical disasters occur due to leakage, explosions, or uncontrolled reactions in industries.
2. Causes include poor maintenance, human error, faulty equipment, and lack of safety audits.
3. Toxic gas exposure causes respiratory failure, blindness, burns, and death.
4. Chemical spills contaminate water bodies and soil.
5. Long-term health effects include cancer, neurological damage, and birth defects.
6. Industries must follow chemical storage and handling norms (MSDS).
7. Emergency plans like isolation zones and immediate evacuation are essential.
8. On-site and off-site DM plans prevent escalation of incidents.

- Sensors and alarms must monitor temperature, pressure, and gas leaks.
- Training workers and conducting mock drills reduce risk.

CaseStudy:BhopalGasTragedy(1984)

Leakage of MIC gas killed over 3,000 instantly and affected 5 lakh people long-term—world's worst industrial disaster.

17. Building and Coal Mine Fires – Causes, Prevention & Control – 10 Points

- Building fires arise from electrical faults, gas leaks, kitchen accidents, and flammable materials.
- Coal mine fires occur due to methane ignition, coal dust explosions, or spontaneous heating.
- Poor ventilation accelerates fire spread and suffocation deaths.
- High-rise buildings without proper exits result in mass casualties.
- Smoke inhalation is the major cause of deaths in building fires.
- Fire safety norms include extinguishers, alarms, sprinklers, and fire-resistant materials.
- Regular electrical inspections prevent short circuits.
- Coal mine fires require sealing, cooling, or flooding techniques to control.
- Emergency evacuation routes must be clearly marked.
- Fire department and trained staff improve survival rates.

CaseStudy:KolkataAMRIHospitalFire(2011)

Short circuit triggered a fire in basement; 92 patients died—highlighted failure of fire exits and safety negligence.

18. Forest Fires & Oil Spills – Environmental & Economic Impact – 10 Points

Forest Fires

- Caused by lightning, heatwaves, human negligence, or intentional burning.
- Destroy forests, wildlife, and natural habitats.
- Release smoke and CO₂, worsening air pollution and climate change.
- Lead to soil erosion and loss of biodiversity.

Oil Spills

- Oil spills occur due to tanker accidents, pipeline leaks, or offshore drilling failures.
- Oil creates a toxic film that kills fish, birds, and marine mammals.
- Coastal tourism and fisheries suffer severe financial losses.
- Cleanup is expensive and technically challenging.
- Bioremediation and mechanical skimmers help reduce pollution.
- Strict monitoring of ships and pipelines prevents spills.

Case-Study:

Deepwater Horizon Oil Spill (2010) in the Gulf of Mexico spilled 4.9 million barrels of oil—largest marine oil disaster.

19. Air and Water Pollution – Sources, Effects & Control Measures – 10 Points

- Air pollution sources include vehicles, industries, construction, and biomass burning.
- Water pollution sources include sewage discharge, industrial effluents, pesticides, and plastic waste.
- Air pollution causes asthma, lung cancer, heart disease, and reduced visibility.
- Water pollution leads to cholera, typhoid, jaundice, and aquatic death.
- Industrial areas suffer from smog and acid rain.
- Wastewater treatment plants help remove contaminants.
- Switching to clean fuels (CNG, EVs) reduces air pollution.
- Rainwater harvesting reduces wastewater pressure.
- Strict PCB norms and environmental laws prevent industrial pollution.
- Community awareness and solid waste management are key to pollution control.

Case-Study:

Delhi Air Pollution Crisis (2019) – AQI crossed 500; schools were closed; major health emergency declared.

20. Deforestation – Causes & Consequences – 10 Points

- Causes include agriculture expansion, logging, mining, and urbanization.
- Cutting forests increases carbon emissions and global warming.
- Loss of habitat threatens wildlife and leads to species extinction.
- Soil erosion increases as tree roots no longer hold soil.
- Rainfall patterns change, contributing to drought and floods.
- Water cycle is disrupted, reducing groundwater recharge.
- Indigenous communities lose homes and livelihood.
- Deforestation accelerates climate change.

- Reforestation and afforestation restore ecosystem balance.
- Laws like Forest Conservation Act help protect forests.

Case-Study:

Amazon Rainforest deforestation (Brazil) is leading to massive biodiversity loss and global climate impacts.

21. Epidemics & Pandemics – Case Studies & Response Strategies – 10 Points

- Epidemics are sudden outbreaks of diseases within a region.
- Pandemics spread globally affecting multiple countries.
- Causes include viruses, bacteria, poor sanitation, and population density.
- Major impacts include mass illness, death, and economic shutdown.
- Governments must implement surveillance and rapid testing.
- Quarantine, isolation, and contact tracing reduce spread.
- Hospitals must be equipped with PPE, ventilators, and ICU beds.
- Public awareness and hygiene practices reduce infection.
- Vaccination is key to long-term control.
- International cooperation is essential to share data and medicine.

CaseStudy:COVID-19Pandemic(2020–22)

India imposed nationwide lockdown, used DM Act 2005, expanded hospital capacity, and vaccinated millions.

UNIT-IV-DISASTER PREPAREDNESS PLAN

22. Disaster Preparedness Plans – Importance & Scope

- A Disaster Preparedness Plan outlines actions to be taken before, during, and after a disaster.
- Its main goal is to reduce loss of life, property damage, and response time.
- It enhances community resilience through awareness, training, and readiness.
- Preparedness ensures faster rescue, evacuation, and medical support.
- It builds coordination between government, NGOs, military, and local bodies.
- Early warning systems become effective only when preparedness exists.
- Helps allocate resources like shelters, medicines, boats, and rescue teams.
- Ensures continuity of essential services such as water, power, and communication.
- Reduces financial burden during disasters through planned mitigation.
- Scope includes all hazards—natural, man-made, biological, and technological.

Case-Study:

Odisha's Cyclone Preparedness Plan saved lakhs of lives during **Cyclone Phailin (2013)** by evacuating 1 million people.

23. Essentials of a Disaster Preparedness Plan & Risk Management – 10 Points

- Risk assessment** identifies vulnerable areas, populations, and hazards.
- Resource mapping** lists available equipment, shelters, volunteers, and vehicles.
- A clear **communication plan** ensures warnings reach people immediately.
- Evacuation routes** must be marked and practiced regularly.
- Emergency shelters** should include food, water, beds, and medical supplies.
- Search and rescue teams** must be trained and ready.
- Risk management includes mitigation like retrofitting and flood barriers.
- Medical preparedness** includes first-aid kits, ambulances, and hospital readiness.
- Mock drills test the efficiency of the preparedness system.
- Plans must be updated regularly based on new threats and past experiences.

Case-Study:

Japan's earthquake preparedness includes drills, strong building codes, and mass communication systems—reducing deaths in major quakes.

24. Community-Based Disaster Preparedness Plans – 10 Points

- Communities are the first responders during any disaster.
- Local knowledge helps identify hazard-prone locations and risks.
- Community volunteers assist in evacuation and first aid.
- Training programs teach emergency skills like CPR and fire safety.
- Village committees help in resource distribution during emergencies.
- Mock drills prepare the community for real disaster situations.
- Community warning systems (sirens, WhatsApp groups) reduce response time.
- Involving local schools strengthens preparedness for children.

- Self-help groups assist in managing shelters and food distribution.
- Strong community networks speed up post-disaster recovery.

Case-Study:

In the **Tamil Nadu Tsunami (2004)**, villages with trained community groups evacuated faster and had fewer casualties.

25. Disaster Prediction & Early Warning Systems – 10 Points

- Early warning systems use science and technology to detect approaching hazards.
- IMD predicts cyclones using satellites, radars, and ocean buoys.
- ISRO satellites monitor floods, drought indicators, and forest fires.
- Seismographs detect tremors and help forecast aftershocks.
- Warning messages are sent through SMS, TV, radio, and mobile alerts.
- IoT sensors monitor river water levels and dam pressure.
- Early warnings reduce casualties by allowing time to evacuate.
- Data-based prediction helps government prepare rescue teams earlier.
- AI models analyze climate patterns to predict extreme weather.
- Prediction systems require regular maintenance and global data sharing.

Case-Study:

Before **Cyclone Amphan (2020)** IMD's accurate early warning helped evacuate over 3 lakh people from West Bengal.

26. Safety Measures & Response Strategies for Various Disasters – 10 Points

- Earthquake:** Drop–Cover–Hold, evacuate open areas, avoid elevators.
- Flood:** Move to higher ground, switch off electricity, avoid bridges.
- Cyclone:** Secure roofs, evacuate coastal areas, stock emergency supplies.
- Drought:** Store water, use drip irrigation, protect food stock.
- Fire:** Stop–Drop–Roll, use fire extinguishers, avoid smoke-filled rooms.
- Chemical Leak:** Cover nose with cloth, evacuate windward side.
- Tsunami:** Immediately move 3 km inland or 30 m above sea level.
- Pandemic:** Wear masks, isolate infected persons, maintain hygiene.
- Landslides:** Avoid steep slopes, listen to geological warnings.
- Immediate response must follow the 3 E's: Evacuation, Emergency care, and Essential supplies.

Case-Study:

During **Cyclone Fani (2019)**, Odisha's response system—shelters, boats, volunteers—controlled the death rate to below 50.

27. Role of Information, Education & Training in Disaster Preparedness – 10 Points

- Education spreads awareness about how to act during disasters.
- School curriculum includes DM chapters to train students.
- Posters, drills, and campaigns remind people of safety measures.
- Training programs teach communities CPR, first-aid, and search operations.
- Public workshops improve knowledge of risks in local regions.
- Social media helps spread crucial warnings rapidly.
- Training reduces panic and increases confidence during emergencies.
- Simulations prepare government officials and response teams.
- Community volunteers learn to coordinate shelters and relief.
- Overall, education reduces vulnerability and strengthens resilience.

Case-Study:

Japan conducts nationwide earthquake drills every year on September 1 (Disaster Prevention Day).

28. Government & NGO Involvement in Disaster Management – 10 Points

- Government forms policies, directs agencies, and allocates funds.
- NDMA, SDMA, and DDMA coordinate disaster planning.
- NDRF, Police, and Army handle rescue and relief operations.
- NGOs like Red Cross provide medical aid, food, and shelter.
- NGOs train volunteers and run community preparedness programs.
- Government and NGOs collaborate for long-term rehabilitation.
- Local bodies manage crowd control, evacuation, and shelters.
- SHGs and charitable groups help distribute food and water.
- International NGOs support with equipment and technical assistance.

- Combined efforts ensure faster recovery and minimal casualties.

Case-Study:

During the **2013 Uttarakhand floods**, NDRF, Army, Red Cross, and NGOs jointly rescued 1 lakh+ stranded tourists and residents.

29. Role of IT in Disaster Management (Satellite Data, IoT, GIS) – 10 Points

- Satellite images help monitor cyclones, rainfall patterns, and forest fires.
- GIS mapping identifies vulnerable zones and disaster hotspots.
- IoT sensors monitor river water levels, landslide movement, and air quality.
- Drones assist in aerial survey for search and rescue.
- Big data and AI help predict hurricanes and drought patterns.
- Weather radars track storms in real-time.
- Apps send early warnings to citizens instantly (NDMA, IMD apps).
- Cloud-based systems ensure communication among agencies.
- Remote sensing helps assess post-disaster damage.
- Technology shortens response time and enhances accuracy.

Case-Study:

ISRO's satellite-based monitoring during the **Kerala Floods 2018** helped officials identify flooded regions and plan rescues.

UNIT V – CASE STUDY WITH SURVEY

30. Methodology for Conducting Disaster Case Studies

- Problem Identification:** Clearly identify the disaster type and affected region.
- Objective Setting:** Define what you want to study—causes, impacts, response, recovery.
- Data Collection:** Collect primary data (interviews, surveys, field visits) and secondary data (government reports, satellite images).
- Timeline Analysis:** Prepare a chronological sequence of events before, during, and after the disaster.
- Hazard Mapping:** Use maps, GIS, and satellite data to locate vulnerable areas.
- Impact Assessment:** Evaluate social, economic, environmental, and infrastructural damage.
- Response Evaluation:** Analyze actions taken by government, NDRF, NGOs, and communities.
- Gap Identification:** Identify failure areas—late warnings, weak infrastructure, lack of preparedness.
- Recommendations:** Suggest improvements based on global best practices.
- Report Writing:** Present structured findings with charts, tables, photographs, and maps.

Example:

Methodology applied in Kerala Floods 2018 case study involves demographic survey, rainfall data, dam discharge records, and ISRO satellite mapping.

31. Analysis of Disaster Impact & Recovery Processes – 10 Points

- Social Impact:** Loss of life, displacement, psychological trauma, disruption of education.
- Economic Impact:** Damage to crops, businesses, roads, transport, and public infrastructure.
- Environmental Impact:** Soil erosion, landslides, contamination, habitat loss.
- Health Impact:** Spread of diseases, injuries, lack of sanitation.
- Immediate Response:** Search and rescue operations, temporary shelters, emergency care.
- Relief Measures:** Distribution of food, water, medicines, and clothing.
- Reconstruction:** Rebuilding homes, schools, bridges, and community structures.
- Livelihood Restoration:** Financial compensation, crop insurance, job programs.
- Psychosocial Support:** Counseling services for affected families.
- Long-term Recovery:** Implementation of mitigation measures to prevent future disasters.

Case-Study:

After the **2015 Nepal Earthquake**, recovery included rebuilding homes with earthquake-resistant designs and foreign aid support from 50+ countries.

32. Best Practices in Disaster Management from Global Case Studies – 10 Points

- Japan:** Strict earthquake-resistant building codes reduce structural collapse.
- USA (Hurricane Management):** Advanced hurricane tracking systems and FEMA coordination.
- Netherlands:** World's best flood control using dikes, storm surge barriers, and pumps.
- Bangladesh:** Cyclone shelters and community warning systems reduced deaths dramatically.
- Australia:** Bushfire early warning apps and satellite fire detection.
- India:** Odisha's cyclone evacuation model recognized by UN.

7. **Philippines:** Community-based disaster preparedness and barangay-level action teams.
8. **Chile:** Fast earthquake response through national rapid action teams.
9. **South Korea:** Nationwide earthquake and fire drills conducted annually.
10. **Indonesia:** Tsunami monitoring using deep-ocean pressure sensors (DART buoys).

Conclusion:

Global best practices improve policy-making and enhance preparedness in disaster-prone countries.

33. Flood Control – 10 Points

1. Construction of dams and reservoirs to regulate river flow.
2. Embankments and levees along rivers prevent overflow.
3. Watershed management increases ground absorption and reduces runoff.
4. Urban drainage improvement prevents waterlogging.
5. Floodplain zoning restricts construction in high-risk areas.
6. Dredging increases river depth to hold more water.
7. Rainwater harvesting reduces pressure on drainage systems.
8. Afforestation increases water retention in soil.
9. Early warning systems ensure timely evacuation.
10. Post-flood measures include water purification and vector control.

Case-Study:

The Netherlands uses storm surge barriers and dikes to prevent flooding, setting the world's best example of flood control engineering.

34. Earthquake – 10 Points

1. Earthquakes are sudden vibrations caused by tectonic plate movement.
2. Tsunamis may accompany undersea earthquakes.
3. Effects include building collapse, ground rupture, and landslides.
4. Earthquake-resistant building design reduces casualties.
5. Retrofitting old structures increases safety.
6. Seismic zoning maps guide safe construction practices.
7. Public training in “Drop–Cover–Hold” saves lives during shaking.
8. Emergency kits with water, food, and medical supplies are essential.
9. Early warning technology (P-wave detection) helps prepare seconds early.
10. Post-earthquake recovery involves restoring utilities, communication, and shelters.

Case-Study:

Japan Earthquake 2011: Massive tsunami damage, but infrastructure damage was minimized due to strict building codes.

35. Nuclear Disaster – 10 Points

1. Nuclear disasters occur due to meltdown, explosion, cooling failure, or natural hazards.
2. Major hazards include radiation release and long-term environmental contamination.
3. Health impacts include cancer, genetic damage, and acute radiation syndrome.
4. Large exclusion zones may remain uninhabitable for decades.
5. Emergency shutdown systems (SCRAM) reduce risk.
6. Radiation suits and dosimeters protect responders.
7. Evacuation plans and shelter-in-place protocols protect citizens.
8. Continuous radiation monitoring is essential around nuclear plants.
9. International Atomic Energy Agency (IAEA) sets global safety norms.
10. Proper waste disposal prevents long-term contamination.

Case-Study:

Fukushima Nuclear Disaster (2011) caused by tsunami—reactor meltdown forced 1.5 lakh people to evacuate.

36. Chemical Disaster – 10 Points

1. Chemical disasters occur due to toxic leaks, explosions, or industrial accidents.
2. Causes include poor maintenance, human error, operational failure, and unsafe storage.
3. Impacts: respiratory failure, poisoning, burns, environmental contamination.
4. On-site emergency plans guide workers during accidents.
5. Off-site emergency plans protect the public through alerts and evacuation.
6. Industries must follow Material Safety Data Sheets (MSDS) for chemical handling.
7. Gas detectors and alarms help in early leak detection.
8. Regular audits and safety inspections reduce accident probability.
9. Training workers in PPE usage increases safety.
10. Public awareness and disaster drills minimize casualties.

Case-Study:

Visakhapatnam LG Polymers Gas Leak (2020) released styrene gas, causing deaths and widespread health impacts